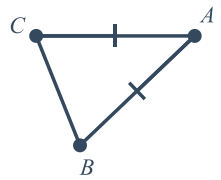
Line segments and distance

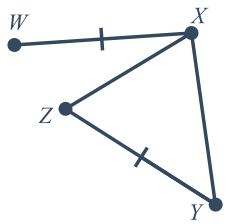
1 Draw the diagram that shows $\overline{EG}$ and a point F that is not on $\overline{EG}$.

2 The diagram shown has two congruent segments marked:



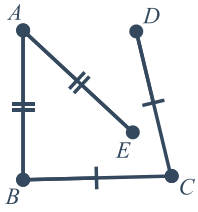
Identify which segment is equal in length to $\overline{AB}$.

3 The diagram shown has two congruent segments marked:



Identify which segment is equal in length to $\overline{ZY}$.

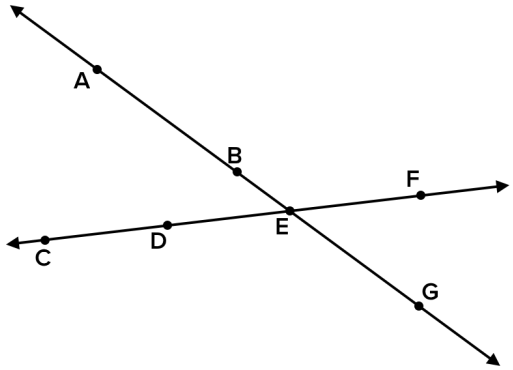
4 The diagram shown has two pairs of congruent segments marked:



Identify which segment is equal in length to $\overline{AB}$.

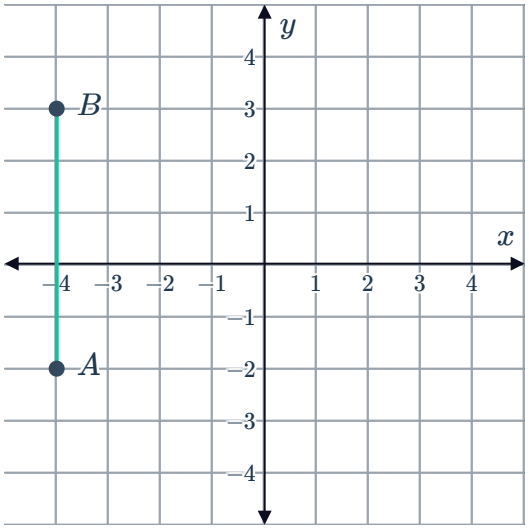
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5 What is another way to describe the exact same set of points as $\overline{FD}$?

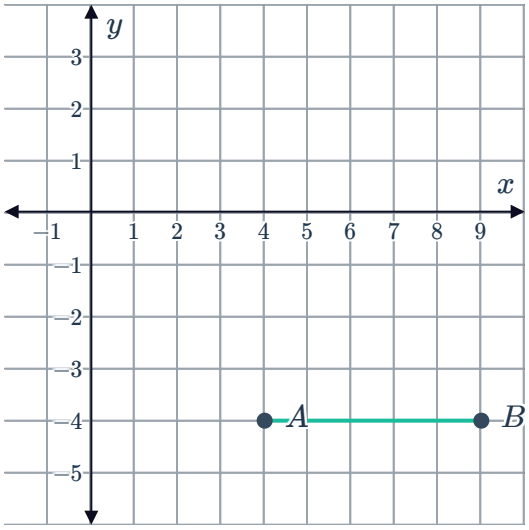


6 Find the length of the indicated segment.

a The vertical segment joining $A\,(-4,-2)$ and $B\,(-4,3)$



b The horizontal segment joining $A\,(4,-4)$ and $B\,(9,-4)$



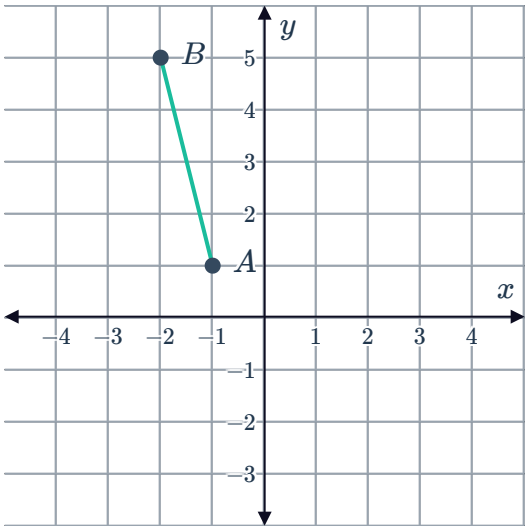
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7 Consider $\overline{AB}$ on the graph.

a Identify the vertical component of the distance from A to B.

b Identify the horizontal component of the distance from A to B.

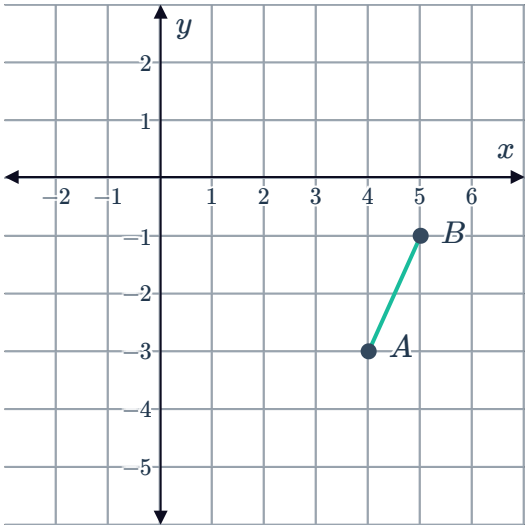
c If the length of $\overline{AB}$ is c, find the exact value of c.



8 Consider $\overline{AB}$ on the graph.

a Identify the vertical component of the distance from A to B.

b Identify the horizontal component of the distance from A to B.



9 State the formula for finding the distance d between two points (x_1, y_1) and (x_2, y_2).

10 Find the missing values in the following working that is used to find the distance between points $(-5, 10)$ and $(4, -5)$.

Distance $= \sqrt{(-5 - \square)^2 + (10 - \square)^2}$

11 Find the distance between $A\,(1, 4)$ and $B\,(7, 12)$, rounding your answer to two decimal places.

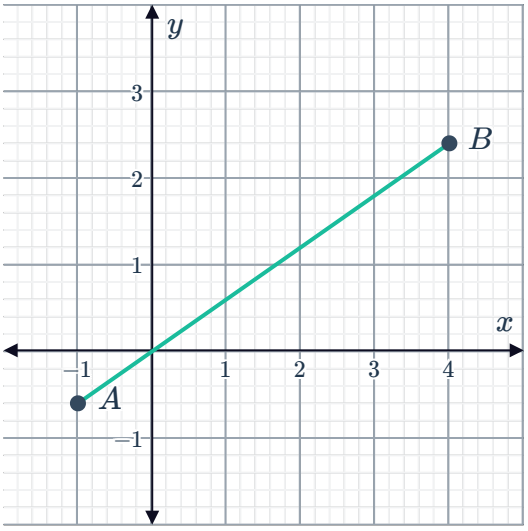
12 Find the exact distance between the following points.

a $A\,(4, 2)$ and $B\,(-8, -7)$

b $A\,(-1, 9)$ and $B\,(-4, 1)$

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13 Find the exact distance from $A\left(-1, -\frac{3}{5}\right)$ to $B\left(4, \frac{12}{5}\right)$.



14 Given that $M\,(-9, -1)$ and $N\,(1, 5)$.

- **a** Find the exact distance from M to the origin.
- **b** Find the exact distance from N to the origin.
- **c** Which point is closer to the origin?

15 Given that $P\,(5, -1)$, $M\,(10, -6)$, and $N\,(1, 2)$.

- **a** Find the exact distance from P to M.
- **b** Find the distance from P to N.
- **c** Which point is further from P?

16 The points $M\,(0, -8)$ and $R\,(6, -2)$ are the end points of a segment. $N\,(5, -3)$ is a point on this segment.

- **a** Find the exact distance between M and N.
- **b** Find the exact distance between N and R.
- **c** In what ratio does N divide $\overline{MR}$?

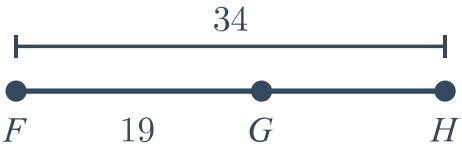
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17 Find the length of the indicated segment.

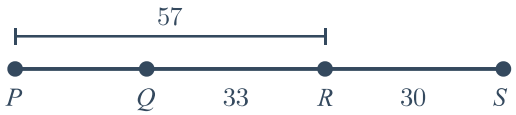
a $\overline{CE}$



b $\overline{GH}$



18 Consider the following figure.



a Find QS.

b Find PQ.

19 In the diagram, $JK = 5x + 2$, $KL = 7x + 4$, and $JL = 42$.



a Find x.

b Find KL.

20 In the diagram, $JT = 5x + 5$, $CT = 76$, and $CJ = 4x + 8$. Find JT.



21 For each of the following scenarios:

i Find x.

ii Find CB.

a Suppose the point B lies on $\overline{CD}$ between the points C and D, and $CB = 2x$, $BD = 4x$, and $BD = 12$.

b Suppose the point B lies on $\overline{CD}$ between the points C and D, and $CB = 3x - 6$, $BD = 4x + 3$, and $CD = 18$.

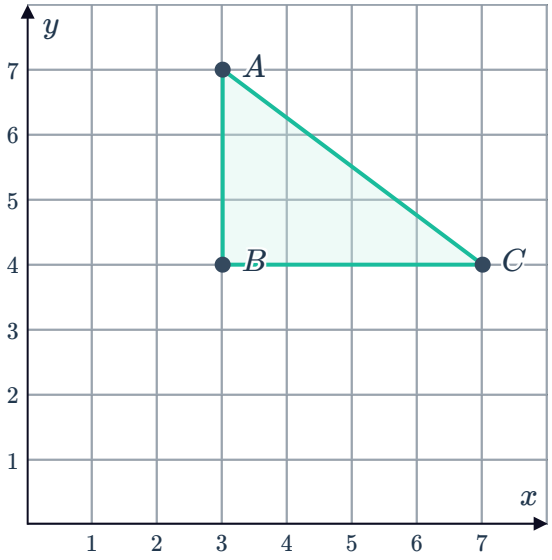
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Additional questions

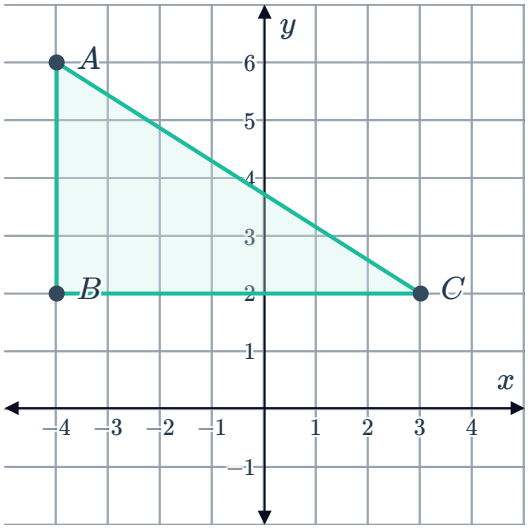
22 For each of the following right triangles:

- **i** Determine AB.
- **ii** Determine BC.
- **iii** Using the Pythagorean theorem, calculate AC, rounding your answer to two decimal places if necessary.

a



b



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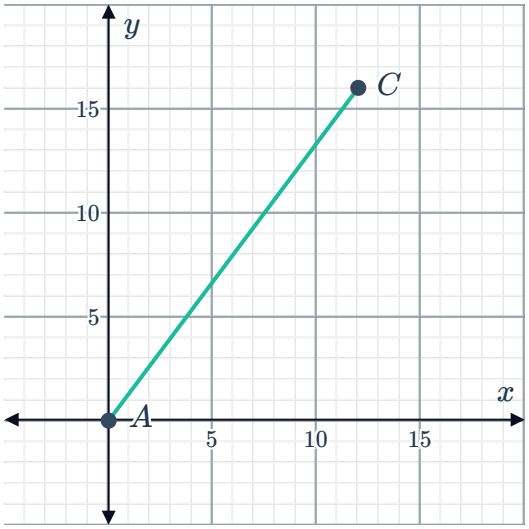
23 For each of the following coordinate planes:

i Copy the coordinate plane and draw a right triangle with $\overline{AC}$ as the hypotenuse.

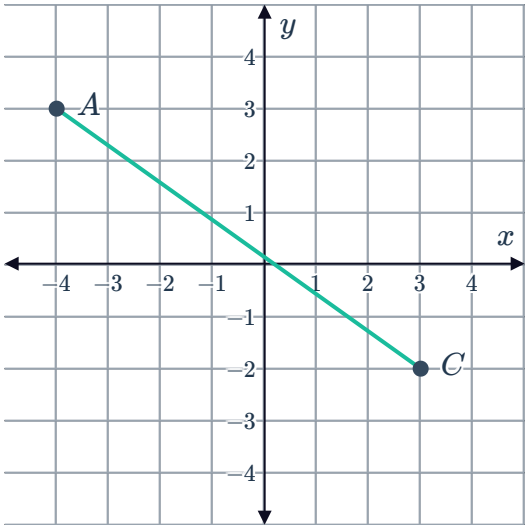
ii Label the shorter two sides with the measure of their lengths.

iii Use the Pythagorean theorem to find the distance between A and C, round your answer to one decimal place if necessary.

a $A = (0, 0), C = (12, 16)$

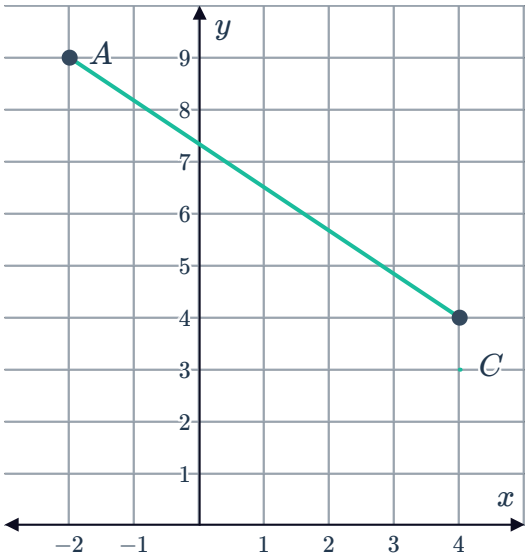


b $A = (-4, -3), C = (3, -2)$

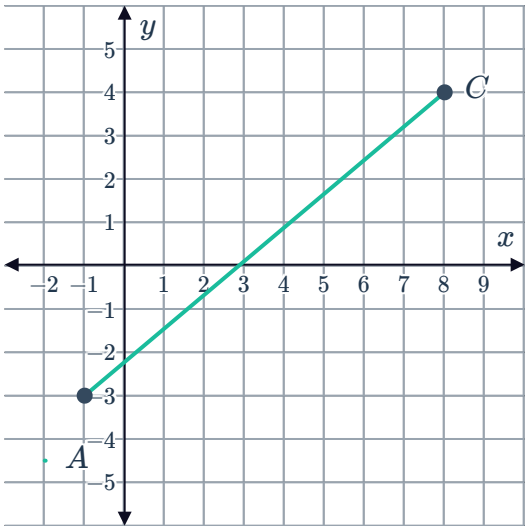


24 Consider $\overline{AC}$ that has been graphed on the coordinate plane. Find AC, rounding your answer to two decimal places.

a



b



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25 Find the distance between Point A and Point B, rounding your answer to two decimal places if necessary.

a $A\,(1,4)$ and $B\,(7,12)$

b $A\,(4,2)$ and $B\,(-8,-7)$

c $A\,(-1,9)$ and $B\,(-4,1)$

d $A\left(-1,-\frac{3}{5}\right)$ and $B\left(4,\frac{12}{5}\right)$

26 Given $P\,(4,3)$, $M\,(-3,-4)$ and $N\,(-7,1)$:

a Find the distance from P to M, rounding your answer to two decimal places.

b Find the distance from P to N, rounding your answer to two decimal places.

c Determine the point that is further from P.

27 A circle with center at point $C\,(3,4)$ has point $A\,(-12,12)$ lying on its circumference. Find the radius of the circle.

28 $ABCD$ is a rhombus whose vertices are $A\,(1,2)$, $B\,(3,10)$, $C\,(11,12)$ and $D\,(9,4)$. Find the exact length of the diagonals:

a $\overline{AC}$

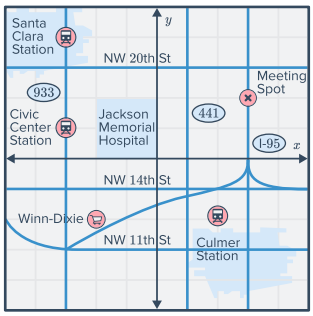
b $\overline{BD}$

29 Mateo is meeting his mom at the Meeting Spot shown on the map below after a doctor's appointment. They need to catch the train home.

a Which of the three train stations is the closest to their meeting spot?

b What might not make the closest train station the best station to go to in reality?

c If they needed to stop at the Winn-Dixie before going home, which station would have the shortest trip distance?



30 $\overline{AM}$ is a vertical line segment which is 3 units long. If A is the point $(-2,6)$:

a How many possible coordinates of M are there?

b What are the possible coordinates of M?

c How would your answers to parts (a) and (b) change if we were not told the line segment was vertical?

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32 Dayanna and David were asked for the formula for the distance between two points $A(x_1, y_1)$ and $B(x_2, y_2)$ on the coordinate plane.

Dayanna wrote $AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$.

David wrote $AB = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$.

Who is correct? Explain your answer.

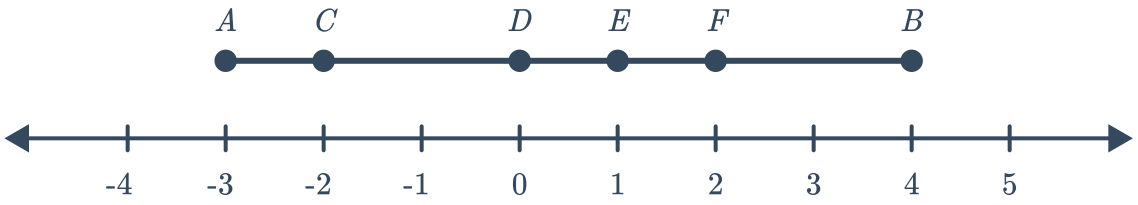
33 Fyfe is asked to find the length of the line segment $\overline{AB}$ with endpoints $A(-4, 7)$ and $B(3, -5)$. Below is their solution.

1	$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$	Distance formula
2	$= \sqrt{(3 - -5)^2 + (-4 - 7)^2}$	Substitution Property of Equality
3	$= \sqrt{(8)^2 + (-11)^2}$	Simplify
4	$= \sqrt{64 + 121}$	Simplify
5	$= \sqrt{185}$	Simplify

a Identify the step where Fyfe makes an error and explain their misunderstanding.

b Correct Fyfe's work.

34 Use the diagram to find each segment length:



a AC

b AE

c EF

d FB

e DB

f AB

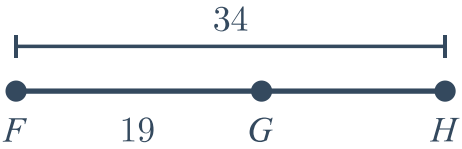
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35 Write an equivalence statement using the segment addition postulate, then find the length of indicated segment.

a CE

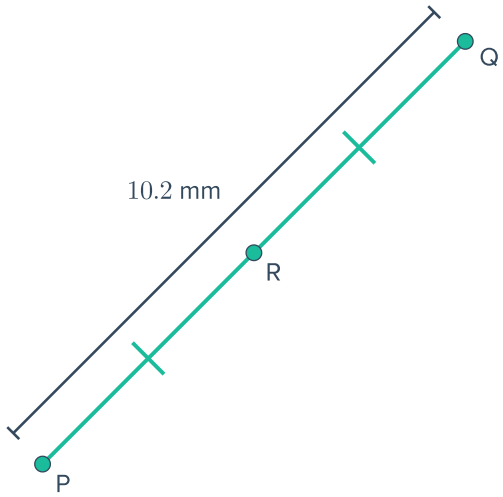


b GH

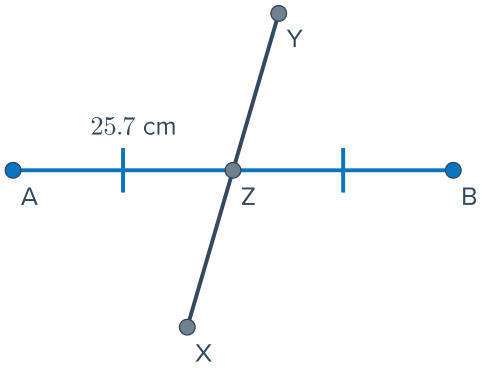


36 Use the diagrams provided to find the indicated length:

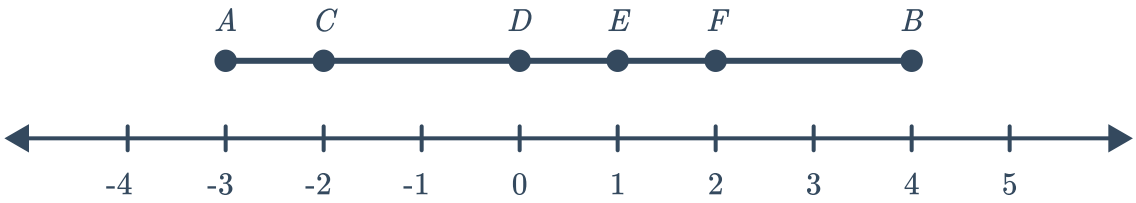
a PR



b AB



37 Consider the segment shown in the diagram:



a Which segments are congruent to $\overline{AC}$?

b Which segments are congruent to $\overline{AE}$?

c If a point G was added at -1, how many additional congruent segments would you find in parts (a) and (b)?

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38 Use the segment addition postulate to find the unknown length.

a Given: $JK = 5x + 2$, $KL = 7x + 4$ and $JL = 42$. Find KL.



b Given: $JT = 5x + 5$, $CT = 76$ and $CJ = 4x + 8$. Find JT.



39 Suppose the point B lies on the segment $\overline{CD}$ between the points C and D.

a If $CD = 4x$, $BD = 2x$ and $CB = 12$, what is the length of $\overline{BD}$?

b If $CB = 3x - 6$, $BD = 4x + 3$ and $CD = 18$, what is the length of $\overline{CB}$?

40 In the diagram below, M is the midpoint of $\overline{KL}$. Find the length ML.



41 Given that points X, Y, and Z are collinear, and that $XY = 12$ and $XZ = 18$, what are the possible lengths for YZ?
Draw diagrams that support your solution.

42 Given that $AB = 10$, $BC = 5$, and $AC = 12$ could points A, B, and C be collinear? Explain.

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